MCC505 Probability & Statistics, Quiz-II, Monsoon 2024-25 Max Marks 10, Time: **30 min**; Code- $\boxed{A}$

Name.. Adm. No...Sec.................

Put a tick mark (✓) against the correct answer. There is only one correct answer for each question. Each correct answer would fetch 1 mark and no partial credit. Overwriting is not allowed and would attract zero credit.

(1) In a precision bombing attack there is a 50% chance that any one bomb will strike the target. Two direct hits are required to destroy the target completely. The minimum number of bombs must be dropped to give a 99% chance or better of completely destroying the target:
✓(A) 11 (B) 10 (C) 9 (D) 12

(2) For Poisson distribution with unit mean, mean deviation about mean is:
(A) $\frac{1}{e}$ standard deviation
(B) $\frac{e}{2}$ standard deviation
✓(C) $\frac{2}{e}$ standard deviation
(D) e standard deviation

(3) If the probability is 0.40 that a child exposed to a certain disease will contain it, the probability that the tenth child exposed to the disease will be the third to catch it is:
✓(A) 0.0645 (B) 0.070 (C) 0.0570 (D) 0.055

(4) A small voting district has 101 female voters and 95 male voters. A random sample of 10 voters is drawn. The probability that exactly 7 of the voters will be female is:
✓(A) 0.130 (B) 0.115 (C) 0.0120 (D) 0.0140

(5) Suppose that the probability for an applicant for a driver's licence to pass the road test on any given attempt is 2/3. The probability that the applicant will pass the road test on the third attempt is:
✓(A) 0.074 (B) 0.0689 (C) 0.079 (D) 0.0675

(6) Let T be the time of emission of the first electron from the Cathode of a vacuum tube. Under certain physical assumptions, it turns out that T has an exponential distribution with mean $\frac{1}{2}$. Then $P(1 < T < 2)$ is
✓(A) $e^{-2} - e^{-4}$ (B) $e^{-1} - e^{-2}$ (C) $e^{-1} - e^{-4}$ (D) $e^{-2} - e^{-3}$

(7) Suppose the breaking strength of cotton fabric (in pounds), say X, is normally distributed with $E(X) = 165$ and $V(X) = 9$. Assume furthermore that a sample of this fabric is considered to be defective if $X < 162$. What is the probability that a fabric chosen at random will be defective? ($\Phi(z) = P(Z \leq z)$ is the cumulative distribution function of the standard normal distribution).
✓(A) $\Phi(-1)$ (B) $\Phi(1)$ (C) $\Phi(3)$ (D) $\Phi(-3)$

(8) Assume that you have two batteries that have an exponential lifetime with mean 2 hours. As soon as the first battery fails, you replace it with a second battery. What is the probability that the batteries will last at least 4 hours?
(A) $3e^{-3}$ (B) $2e^{-3}$ ✓(C) $3e^{-2}$ (D) $2e^{-2}$

(9) Let X be an exponential random variable with mean parameter one. Then the conditional probability $P(X > 10 | X > 5)$ is equal to
(A) $1 - e^{-5/2}$ (B) $e^{-5/2}$ (C) $1 - e^{-5}$ ✓(D) e^{-5}

(10) The thickness X of a protective coating applied to a conductor designed to work in corrosive conditions follow a uniform distribution over the interval [20, 40] microns. Then mean and standard deviation of X is
(A) $E(X) = 10\ \mu\text{m}, \sigma = \frac{20}{\sqrt{12}}\ \mu\text{m}$
(B) $E(X) = 20\ \mu\text{m}, \sigma = \frac{20}{\sqrt{12}}\ \mu\text{m}$
✓(C) $E(X) = 30\ \mu\text{m}, \sigma = \frac{20}{\sqrt{12}}\ \mu\text{m}$
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MCC505 Probability & Statistics, Quiz-II, Monsoon 2024-25 Max Marks 10, Time: **30 min**; Code- $\boxed{\textbf{B}}$

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MCC505 Probability & Statistics, Quiz-II, Monsoon 2024-25 Max Marks 10, Time: **30 min**; Code-$\boxed{\text{C}}$

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MCC505 Probability & Statistics, Quiz-II, Monsoon 2024-25 Max Marks 10, Time: **30 min**; Code-$\boxed{\text{D}}$

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